

The empirical recurrence for OEIS A229369, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A229369 records “Empirical recurrence of order 50” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 121$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 55$. Its last coefficient is $c_{50} = -48945447305216 \neq 0$, so no further tail satisfies anything shorter; any order-50 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A229369 is “Number of $n \times 5$ 0..2 arrays avoiding 11 horizontally, 22 vertically and 00 antidiagonally.”. It has offset 1 and begins

164, 4144, 122497, 3576912, 104382552, 3043629267, 89084628843, 2599351293506, ...

The entry states:

Empirical recurrence of order 50 (see link above)

As of the “Last modified” line on the live entry (Oct 04 2025 00:09:30, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 121$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 121$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 50: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 50.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 242$ exact terms from $n = 5$ on, it returns order 50 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{50} c_i a(n-i),$$

c_1	2	c_{14}	474802778034481	c_{27}	253574472099344888240	c_{40}	−1255305865556875
c_2	1067	c_{15}	−674381429052674	c_{28}	−304674183093323303470	c_{41}	1082142978239827
c_3	3060	c_{16}	−9376351717040770	c_{29}	−592440183299263476588	c_{42}	−201372934997478
c_4	−356561	c_{17}	9593562844811778	c_{30}	509701869178278662176	c_{43}	−708941586666056
c_5	−980350	c_{18}	136692099463704784	c_{31}	931453512720129553672	c_{44}	3321240184826200
c_6	62769508	c_{19}	−59244827478543374	c_{32}	−763844094652258450348	c_{45}	−55467263355912
c_7	92863746	c_{20}	−1448400006511362121	c_{33}	−988389855427537557168	c_{46}	−16456411310404
c_8	−6633388628	c_{21}	−301413648009542492	c_{34}	976217824811799070352	c_{47}	150100124285009
c_9	−211787424	c_{22}	10654410869688013886	c_{35}	622395401286710926368	c_{48}	39092159575490
c_{10}	430681312292	c_{23}	8572945168408846612	c_{36}	−925234306188649219520	c_{49}	−3444663092510
c_{11}	−451154394184	c_{24}	−50810752839856999610	c_{37}	−87479845095619735360	c_{50}	−489454473052
c_{12}	−17365100195289	c_{25}	−66560257768935084262	c_{38}	539729827222188177632		
c_{13}	25641134019782	c_{26}	148838251548238218172	c_{39}	−178641070472631684352		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 55$, and it is the recurrence OEIS A229369 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{50} - c_1 x^{50-1} - \dots - c_{50}$. Its constant term is $-c_{50} = 48945447305216$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 50 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 50, so $\deg q = 50$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 50, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 50, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -48945447305216 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-50 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A229369.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.